11.5 Gravity in an Accelerating Train

WORKSHEET

The question

A train moves westwards on a straight horizontal track with constant acceleration a, where $a > 0$. Axes are chosen as follows: the origin is fixed in the train; the x-axis is in the direction of the track with the positive x-axis pointing to the East; and the positive y-axis points vertically upwards.

A smooth wire is fixed in the train. It lies in the x-y plane and is bent in the shape given by $ky = x^2$, where k is a positive constant. A small bead is threaded onto the wire. Initially, the bead is held at the origin. It is then released.

(i) Explain why the bead cannot remain stationary relative to the train at the origin.

(ii) Show that, in the subsequent motion, the coordinates (x, y) of the bead satisfy

$$\dot{x}(\ddot{x} - a) + \dot{y}(\ddot{y} + g) = 0$$

and deduce that $\frac{1}{2}(\dot{x}^2 + \dot{y}^2) - ax + gy$ is constant during the motion.

(iii) Find an expression for the maximum vertical displacement, b, of the bead from its initial position in terms of a, k and g.

(iv) Find the value of x for which the speed of the bead relative to the train is greatest and give this maximum speed in terms of a, k and g.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself The train accelerates westwards (in the negative x direction) at a. For an observer in the train, what "fictitious" force appears to act on a mass m?

STAGE 2 An energy equation

Ask yourself Write Newton's law in the train's frame, including the normal reaction $\mathbf{N}$. Then take the dot product with the velocity $(\dot{x}, \dot{y})$. Why does $\mathbf{N}$ disappear?

STAGE 3 How high?

Ask yourself The constant is 0, since the bead starts at rest at the origin. At the highest point the bead is momentarily at rest. What equation does that give?

STAGE 4 The fastest point

Ask yourself Write the speed squared as a function of x alone. Where is it greatest?

Ask yourself Find the gradient of the wire at $x = \frac{ak}{2g}$. How does it relate to the direction of the effective gravity $(a, -g)$?

Try this next

1 **A cannon on a train.** (From STEP I, 2000.) A railway carriage moves north with acceleration a, and a child fires a toy cannon southwards at angle θ and speed V. Show that the range on the carriage floor is greatest when $\tan 2\theta = -\frac{g}{a}$, that this angle is approximately $\frac{\pi}{4} + \frac{a}{2g}$ when a is small, and that the maximum range is then approximately $\frac{V^2}{g} + \frac{V^2 a}{g^2}$.

2 **A headwind.** (From STEP II, 2006.) A projectile is fired north at speed u and angle θ, against a wind that exerts a constant southward force f per unit mass. Find where it lands. Show that if θ exceeds a certain angle α the wind blows it back towards O before it lands, and find α. Describe the motion when $f = g$ and $\theta = 45°$.

3 **Coffee in a train.** Show that the surface of a cup of coffee in the accelerating train is a plane tilted at angle $\arctan(a/g)$ to the horizontal.

4 **Small oscillations.** Find the period of small oscillations of the bead about $x = \frac{ak}{2g}$. (*Hint:* near the new bottom the wire is approximately a circle of radius equal to its radius of curvature.)

Further reading

- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), the chapter on non-inertial frames and fictitious forces.
- **Albert Einstein, *Relativity: The Special and the General Theory*** (various editions). Einstein's own popular account, including the accelerating box.
- **Pedro Ferreira, *The Perfect Theory*** (Little, Brown). A biography of general relativity.

The full guided solution, and what the question leads to at university, is in Section 11.5:
https://beyondstep.keithmcnulty.org/ch11/accelerating-frame.html